Effects of Brightness on Data Identification by AI Models

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Abstract

In today's world of AI, there are many technologies which depend upon AI image recognition, and these depend upon AI's ability to identify objects and patterns in images. Every time the images are taken, they may not have the utmost clarity. Therefore, it is important to investigate how image brightness affects the accuracy of an AI model to identify images, which is also the purpose of this investigation. It was hypothesized that as the brightness level of the image decreases then the accuracy of the AI model decreases. This model used was the VVG16 model developed by Oxford and was tested using images which have brightness levels of –100,-50,0,50,and 100. Then whether the model accurately identified the object was noted and finally the average accuracy was found. Accuracy was 94% when the brightness was 0 or original and lowest of 0% when the brightness was –100 or very minimal. These results show that AI models perform best when images similar to the trained data are the input depicting the importance of brightness in image processing.

Introduction/Background Information

Artificial intelligence, the ability of a machine to think like a human being, has become very prominently used in the past 5 years in image recognition scenarios such as facial recognition to medical diagnosis. The classification models are developed using convolutional neural networks (CNN), which is an algorithm in computers to analyze and interpret data such as images. One well known CNN used for image classification is the VGG16 which was developed by Visual Geometry Group at Oxford University. This CNN consists of 16 layers, so data can be learned efficiently by a computer. One thing to note is that VGG16 works the best at classifying images which are clear.

However, in the real world some images cannot be captured with the utmost clarity, lighting conditions, and brightness. Brightness is a crucial factor as even minute changes can cause the model to misclassify a given image. Research done in the past by many has implied that deep learning models are sensitive to the brightness present in an image. Also, the model has a harder time identifying images with the brightness level which it has not been trained with. In addition, some brightness levels can be positive while other brightness levels are negative. For instance, proper brightness improves the visibility of the object within the image, depicts the object with more vibrant and intense colors, and shows the true colors of the image better. On the other hand, poor brightness can cause a loss of detail in images which are originally dark, causing the already dark areas to turn black in some cases. Nonetheless, models analyze and learn images using gray data or images with no color. Basically, the model turns a image used for training into a gray scale image and then learns that image. Therefore, understanding how brightness levels impact the accuracy of an AI model is crucial due to many technologies being dependent on these AI model's response.

This experiment reveals how changing the brightness level of an image can influence the accuracy of an AI model, VGG16, in this case. This experiment's purpose is to understand the relationship between brightness and accuracy along with how sensitive a model can be along with visualizing how VGG16 classifies data when distorted. This research helps improve future development of various models in the stages of data processing and training. Furthermore, these results would help AI engineers to gain insight on how to train models with a variety of data, so it performs well when unseen data is asked to be classified.

Problem Statement

How does the level of brightness affect the accuracy of an AI image recognition model?

Hypothesis

If the image is more distorted in terms of brightness, then the accuracy of an AI image recognition model will significantly decrease, especially when the brightness is lowered.

Variables

Independent variable: level of brightness

Dependent variable: accuracy of AI image recognition model

Controlled variables:

- Amount of light while taking the photos
- Degree at which photos are taken from
- The distance from the object from which photos are taken
- Device used to take photos

Materials

1) 10 everyday objects

- Pencil
- Composition notebook
- Plastic spoon
- Plant
- Mug
- iPad

- Eyewear
- Water bottle
- Shoes
- Pen

2) Image capturing device

- iPhone 13: to capture photos of all objects listed above

3) Editing software/tool

- iPhone photos app: to adjust brightness

4) AI model coding platforms

- Google Colab: to train model

- Hugging Face: to deploy model

5) Computer or laptop

- To train model and record results

6) Storage device

- USB drive: to safely store photos and have a backup if necessary

7) Stable light source

- Desk lamp or similar light source: to emit consistent lighting when taking photos

8) Prior knowledge of Python

Procedure

1) Create a main folder on the laptop labeled AI Distortion Project

2) Within the main folder, create 10 subfolders each named brightness levels and the object name like pen(-100) and another subfolder named originals for a total of 11 subfolders

3) Next, set up a clear consistent lighting source such as a desk lamp. Make sure lighting, angle the photo is taken, and distance from which photo is taken is the same throughout the data collection.

4) Take 2 clear photos of the following objects (pencil, composition notebook, plastic spoon, plant, mug, iPad, eyewear, water bottle, shoes, and pen) to get a total of 20 images.

5) Save and name each photo like pencil_01.jpg, pencil_02.jpg...pen_02.jpg and input all these into an Excel and save it to the originals folder.

6) Open each original images on the iPhone 13 and use the phone editor to create 5 brightness levels for each photo and save all of these to the brightness folder of the respective object creating a total of 100 images.

1st level – brightness of –100

2nd level - brightness of –50

3rd level – brightness of 0 (original)

4the level – brightness of 50

5th level – brightness of 100

7) Data augmentation is required to code the images to get 5 images of each original using rotations and reflections to get a total of 60 images to develop the model.

8) Open Google Colab and start pre-processing the data, loading the libraries, and training the model of VGG16 base, VGG16 base + FFNN, and VGG16 base + FFNN + Data augugmentation

(Adjusting images)

```
images_decreased = []
height = 64
width = 64
dimensions = (width, height)
for i in range(len(images)):
images_decreased.append( cv2.resize(images[i], dimensions,
interpolation=cv2.INTER_LINEAR))
```

(Loading the three versions of VGG16)

```
vgg_model = VGG16(weights='imagenet',include_top=False,input_shape=(64,64,3))
vgg_model.summary()
```

(Uploading backend files during deployment)

```
# Upload backend files to the dedicated backend space
login(token=access_key)
api = HfApi()

print(f"Uploading backend files to {backend_repo_id}...")
```

```python
api.upload_folder(
folder_path="backend_files",
repo_id=backend_repo_id,
repo_type="space",
)
print(f"Backend files uploaded to {backend_repo_id}.")
```

9) Deploy the model using Hugging Face

10) Input the images of the brightness distorted images and note whether the model classifies it correctly or not

11) Calculate the accuracy of the model using the formula

___________ (correctly identified) _____________

(Total number of trials at the respective brightness level for that object

12) Record all the results in data tables

13) Save the model and back up all images in USB drive for save storage

Data and Results

Object: pencil

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	Yes	No	Yes	Yes
Trial 2	No	Yes	Yes	Yes	Yes
Trial 3	No	Yes	Yes	Yes	No

Trial 4	No	No	Yes	Yes	Yes
Trial 5	No	Yes	Yes	Yes	Yes
Trial 6	No	Yes	No	No	No
Trial 7	No	No	Yes	Yes	Yes
Trial 8	No	Yes	Yes	Yes	Yes
Trial 9	No	Yes	Yes	Yes	Yes
Trial 10	No	Yes	Yes	Yes	Yes
Accuracy	0%	80%	80%	90%	80%

Object: composition notebook

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	No	Yes	Yes	Yes
Trial 2	No	Yes	Yes	Yes	Yes
Trial 3	No	Yes	Yes	Yes	Yes
Trial 4	No	No	Yes	Yes	Yes
Trial 5	No	Yes	Yes	Yes	No
Trial 6	No	Yes	Yes	Yes	Yes
Trial 7	No	No	Yes	Yes	Yes
Trial 8	No	Yes	Yes	Yes	Yes
Trial 9	No	No	Yes	Yes	Yes
Trial 10	No	Yes	Yes	Yes	Yes

Accuracy	0%	60%	100%	100%	90%

Object: plastic spoon

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	No	Yes	Yes	Yes
Trial 2	No	Yes	Yes	Yes	No
Trial 3	No	Yes	Yes	No	Yes
Trial 4	No	No	Yes	Yes	Yes
Trial 5	No	Yes	Yes	Yes	Yes
Trial 6	No	Yes	Yes	Yes	Yes
Trial 7	No	No	Yes	No	Yes
Trial 8	No	Yes	Yes	Yes	Yes
Trial 9	No	Yes	Yes	Yes	Yes
Trial 10	No	Yes	Yes	Yes	Yes
Accuracy	0%	70%	100%	80%	90%

Object: plant

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	No	Yes	Yes	No

Trial 2	No	Yes	Yes	Yes	Yes
Trial 3	No	Yes	Yes	Yes	Yes
Trial 4	No	No	Yes	Yes	Yes
Trial 5	No	Yes	Yes	Yes	No
Trial 6	No	Yes	Yes	Yes	Yes
Trial 7	No	No	Yes	Yes	Yes
Trial 8	No	Yes	Yes	Yes	Yes
Trial 9	No	No	No	Yes	Yes
Trial 10	No	Yes	Yes	No	Yes
Accuracy	0%	60%	90%	90%	80%

Object: mug

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	Yes	Yes	Yes	Yes
Trial 2	No	Yes	Yes	No	Yes
Trial 3	No	Yes	No	Yes	Yes
Trial 4	No	No	Yes	Yes	Yes
Trial 5	No	Yes	Yes	Yes	Yes
Trial 6	No	Yes	Yes	Yes	Yes
Trial 7	No	No	Yes	Yes	No
Trial 8	No	Yes	Yes	Yes	Yes

Trial 9	No	No	Yes	Yes	Yes
Trial 10	No	Yes	Yes	Yes	No
Accuracy	0%	70%	90%	90%	80%

Object: iPad

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	No	Yes	No	Yes
Trial 2	No	Yes	Yes	Yes	Yes
Trial 3	No	Yes	Yes	Yes	Yes
Trial 4	No	No	Yes	Yes	Yes
Trial 5	No	Yes	Yes	Yes	Yes
Trial 6	No	Yes	Yes	Yes	Yes
Trial 7	No	Yes	Yes	Yes	No
Trial 8	No	Yes	Yes	No	Yes
Trial 9	No	Yes	Yes	Yes	Yes
Trial 10	No	No	Yes	Yes	Yes
Accuracy	0%	70%	100%	80%	90%

Object: Eyewear

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	Yes	Yes	Yes	No
Trial 2	No	Yes	Yes	Yes	Yes
Trial 3	No	Yes	No	No	Yes
Trial 4	No	Yes	Yes	Yes	Yes
Trial 5	No	No	Yes	Yes	Yes
Trial 6	No	Yes	Yes	Yes	No
Trial 7	No	Yes	Yes	No	No
Trial 8	No	Yes	Yes	Yes	Yes
Trial 9	No	Yes	Yes	Yes	Yes
Trial 10	No	No	Yes	Yes	Yes
Accuracy	0%	80%	90%	80%	70%

Object: Waterbottle

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	No	Yes	Yes	Yes
Trial 2	No	Yes	Yes	No	Yes
Trial 3	No	Yes	Yes	No	Yes
Trial 4	No	Yes	No	Yes	No
Trial 5	No	Yes	Yes	Yes	Yes

Trial 6	No	Yes	Yes	Yes	Yes
Trial 7	No	Yes	Yes	Yes	Yes
Trial 8	No	Yes	Yes	Yes	Yes
Trial 9	No	No	Yes	Yes	Yes
Trial 10	No	No	Yes	Yes	No
Accuracy	0%	70%	90%	80%	80%

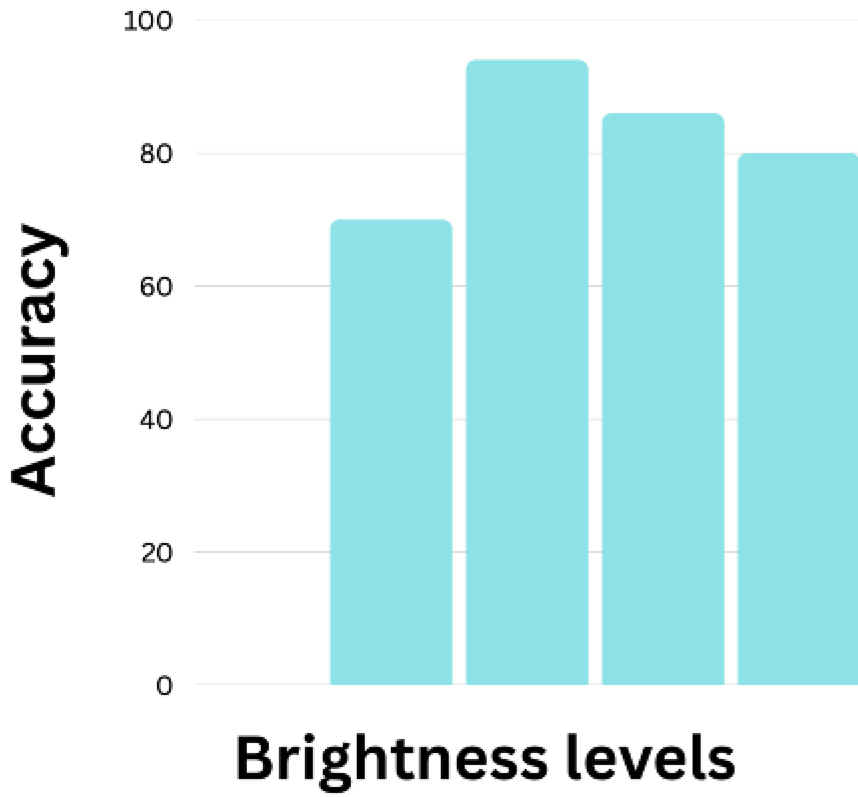
Object: Shoes

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	Yes	Yes	Yes	No
Trial 2	No	No	Yes	Yes	Yes
Trial 3	No	Yes	Yes	Yes	Yes
Trial 4	No	Yes	Yes	Yes	Yes
Trial 5	No	No	Yes	Yes	Yes
Trial 6	No	Yes	Yes	Yes	No
Trial 7	No	Yes	Yes	No	Yes
Trial 8	No	Yes	Yes	No	Yes
Trial 9	No	Yes	Yes	Yes	Yes
Trial 10	No	Yes	Yes	Yes	Yes
Accuracy	0%	80%	100%	80%	80%

Object: Pen

No. Of trials	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Trial 1	No	Yes	Yes	Yes	Yes
Trial 2	No	No	Yes	Yes	No
Trial 3	No	Yes	Yes	No	Yes
Trial 4	No	Yes	Yes	Yes	Yes
Trial 5	No	No	Yes	Yes	Yes
Trial 6	No	Yes	Yes	Yes	Yes
Trial 7	No	No	Yes	Yes	Yes
Trial 8	No	No	Yes	Yes	No
Trial 9	No	Yes	Yes	Yes	Yes
Trial 10	No	Yes	Yes	Yes	Yes
Accuracy	0%	60%	100%	90%	80%

	Brightness level (-100)	Brightness level (-50)	Brightness level (0)	Brightness level (50)	Brightness level (100)
Average Accuracy	0%	70%	94%	86%	80%



Analysis

Based on the graphs and data tables, VGG 16 has no accuracy in predicting images due to the low brightness. However, the accuracy increases significantly for every image when the brightness is –50. When the original image from which the model was trained is inputted, most of the time, i.e. 94% of the time, the model performs the best because there is high accuracy of 94%. The model's accuracy again slightly decreases to 86% when the brightness is increased to 50 and the accuracy again decreases to 80% when the brightness is increased to 100 or the highest amount of brightness is present.

Conclusion

The purpose of this project was to determine how the level of brightness of an image can influence an AI model's accuracy when the model is trained with only original brightness levelled images. The model achieved the highest accuracy of 94% when the objects had the original brightness or the same brightness as the trained images. However, there was a significant drop in accuracy as the brightness level changed. For instance, as soon as the brightness was –50, the accuracy was roughly 70% and when the brightness was +50, the accuracy fell to 86%. To make note of another scenario, the accuracy of the model when the brightness was 100 was 80% while the accuracy when the brightness was 50 was 86%. Since a brightness of 100 means the object was more distorted from the original, the accuracy was lower than the brightness of 50. These results support the hypothesis that if the object was more distorted from the original in terms of brightness, the accuracy did decrease.

These results imply that light and brightness are crucial in image processing because if the brightness varies, then the overall accuracy of that model would decrease. However, accuracy may have been influenced by some external factors which were not controlled properly and that includes the colors of the objects within the images. Some objects were very dark in color while others were very light, which could have worsened when the objects were changed to different bright levels. Darker objects developed almost black regions within the images, and lighter objects developed almost white regions within the images when the brightness levels were changed to –100 and 100, respectively. Nonetheless, on future experiments like this one, the colors of the objects chosen should be the same when addressing this issue. Also, for future experiments, more models should be compared and then the best performing model should be picked and deployed to improve the model's capability to recognize images potentially improving the accuracy.

Understanding how brightness impacts the accuracy of an AI model has a significant impact on many fields such as medicine, self-driving cars, etc. For instance, when one is in the dark and tries to open their phone using the face ID, it wouldn't recognize the image due to minimal lighting. However, if the same person is in a excessively bright area, the phone's ability to recognize that person would be significantly higher. Another example is that medical diagnosis requires normal to greater amounts of brightness to reduce the chances of being misdiagnosed. So, MRI scans or X- ray scans require more light and would result in misdiagnosis or a disease when the scans are captured in very minimal lightning. Misdiagnosis can happen even in ideal condition,but the chances of that would be very minute. These are just a few of the fields in which understanding the relationship between brightness and the accuracy of the AI model. Other fields include self-driving cars, surveillance, etc. Now in all fields listed above, improving the brightness conditions would result in higher accuracy even if the images are taken in differing lighting conditions. To sum it up, better brightness levels can positively impact AI models reliability, safety, and efficiency.

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